

Vishaal Selvaraj

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01 SKILLS

Rust · C++ · Java · Python · Golang · Scala · Dart ·
JavaScript · OpenCL · Bash · Tensorflow · React · Flutter

Cloud · Azure · Kubernetes · Docker · Linux · Networking ·
GPGPU · Backend · System Design · Distributed Systems ·
Frontend · Big Data · Algorithms

Communication · Debugging · Team player

02 EDUCATION

B. TECH
Graduated @ July 2022

Computer Science and Engineering

Graduated with a 9.03 GPA from Vellore Institute of
Technology.

03 EXPERIENCE (4Y+)

MICROSOFT
September 2026 - Current

Senior Software Engineer

- I drove the research, design and development of materialized views and incremental view maintenance within the Spark query engine for Microsoft Fabric, significantly enhancing performance in medallion architecture-based pipelines. This key feature is critical for Microsoft Fabric to compete effectively with its competitors, providing a more robust solution for high-performance data processing.

MICROSOFT
September 2024 - September 2026

Software Engineer - II

- I engineered the support for inplace upgrades in Azure HDInsight on AKS, a unique and competitive feature for any big data PaaS in the market. This capability enabled customers to seamlessly integrate the latest open-source features and fixes into their big data clusters with minimal workload disruption. As a result, many high-revenue enterprise customers (~2 million USD potential revenue) were unblocked during onboarding, allowing them to receive critical updates in a managed and efficient manner.

MICROSOFT
July 2022 - September 2024

Software Engineer - I

- I designed and implemented zone-aware scheduling for Kafka brokers by developing a custom Kubernetes scheduler. This feature was a critical requirement for many key enterprise customers seeking to use

Kafka in HDInsight on AKS. This enabled the onboarding of these key customers by allowing them to create Kafka clusters with zone-resiliency, enhancing the platform's reliability and appeal.

- I wrote several key platform components, including cluster provisioning and gateways, for HDInsight on AKS clusters. These components are crucial to the platform's infrastructure and are expected to meet high reliability standards. I ensured these core components achieved a 99.9% reliability metric, reinforcing the platform's stability and performance.
- I designed and implemented end-to-end provisioning for Ranger clusters in HDInsight on AKS, which play a crucial role in establishing governance for other cluster types such as Trino and Kafka. This feature enabled customers to define row-level policies, topic restrictions, and more, meeting their governance needs. This helped to onboard numerous enterprise customers to the platform by addressing their critical governance requirements.

MICROSOFT
Jan 2022 - July 2022

Software Engineer Intern

- I developed a low-cost alternative serverless scaling infrastructure during my internship and successfully proposed its production deployment when the legacy scaling system crashed. Recognizing this as a critical solution, I presented compelling data from test regions to the leadership team, demonstrating its effectiveness. This initiative prevented significant customer refund losses during the outage and replaced the outdated infrastructure, saving approximately 30,000 USD per region per month in maintenance costs.

SMOKETREES
May 2020 - July 2020

Full-stack development intern

- Worked on building a digital marketplace using Flutter, FastAPI and PostgreSQL for BearLyfe, a Hong Kong based organization that helps users improve their wellbeing.
- Built a salon booking app using Flutter, FastAPI, Firestore and Razorpay for fabsalons, an India based startup that allows users to book salon facilities from their home.